



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

World's smallest, fastest, and longest-range Wi-Fi chip

Morse Micro has launched its second-generation MM8108 system-on-chip (SoC). The company claims the chip achieves data speeds up to 43.33Mbps



due to its sub-GHz 256-QAM modulation across an 8MHz bandwidth. This makes it suitable for various settings,

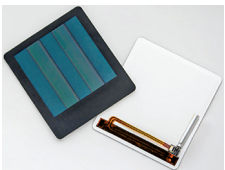
including agriculture, mining, industrial sectors, and urban environments. The SoC features an integrated 26dBm power amplifier and a low-noise amplifier, which help it achieve efficiency and performance, facilitating global compliance without external surface acoustic wave (SAW) filters. This efficiency extends battery life and supports deploying solar-powered Wi-Fi HaLow cameras and IoT devices. It provides a longer range and higher speed than other Wi-Fi HaLow chips, supporting applications.

Morse Micro

<https://www.morsemicro.com>

Self-charging battery for IoT devices

Nichicon and Epishine have launched the SCB-EpNi-2500-000400, a self-charging battery solution. The



company states that the battery provides a power source for small, low-power electronics and is easy

to use. It features solder-free assembly for direct PCB connection, simplifying integration. The battery is designed to

reduce development cycles, decrease setup times, and accelerate the entry of new devices into the market. It supports mass production and efficient deployment, promoting cost savings and scalability for stakeholders.

Nichicon and Epishine

<https://www.epishine.com>

www.nichiconbattery.com

AI chips for robots, AI boxes, and city security

Ambarella has launched the N1-655 edge GenAI system-on-chip (SoC). The company states that the SoC can decode 12 simultaneous 1080p30 video streams



while processing these streams and executing multiple multi-modal vision-language models (VLMs) and traditional CNNs.

It achieves this AI

processing capability while consuming only 20 watts of power—less than cloud processors. The SoC can operate models such as Phi, Gemma, LLaVA-OneVision, and Llama independently of internet connectivity, making it suitable for use in on-premise AI boxes, autonomous mobile robots (AMRs), and smart-city security video recorders. The SoC is ideal for developers and manufacturers in smart city applications. It is designed for sectors prioritising privacy, processing power, and energy efficiency, supporting applications where data is processed locally instead of being sent to the cloud.

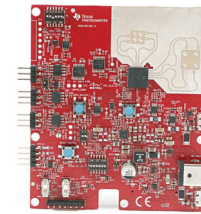
Ambarella

<https://www.ambarella.com>

Industry's first single-chip 60GHz mmWave radar sensor

Texas Instruments (TI) has introduced

the AWRL6844 60GHz mmWave radar sensor that supports occupancy monitoring for seat belt reminders, child presence detection, and intrusion



detection using a single chip and edge AI algorithms to enhance driving safety. The edge AI-enabled AWRL6844 radar sensor allows engineers to com-

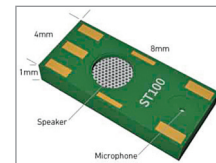
bine multiple in-cabin sensing functions into one device, simplifying design and reducing costs per vehicle. The sensor's functions include occupant detection to activate seat belt reminders, monitoring for unattended children in real time, and scanning to minimise false alarms from intrusion detection systems.

Texas Instruments (TI)

<https://www.ti.com>

World's smallest integrated speaker-microphone

SonicEdge LTD has launched SonicTwin, which is claimed to be the world's smallest integrated speaker-microphone package. SonicTwin is designed for True



Wireless Stereo (TWS) earbuds and headphones, with dimensions of 8mm x 4mm x 1mm. This

device combines sound generation and sensing capabilities, setting a new standard in compact audio technology. Using SonicEdge's patented modulated ultrasound technology, the SonicTwin eliminates traditional mechanical vibrations, enabling the integration of speakers and sensors. The SonicTwin also simplifies the production process by reducing component count, which